

Visualizing the COVID-19 Pandemic: Retrospective Insights and Implications for Public Health Strategies

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Abstract—This paper presents a comprehensive visualization-driven data analysis of the COVID-19 pandemic, focusing on key aspects such as new cases, vaccination rates, and demographic factors. Various visualization techniques, including line plots, scatter plots, pie charts, and bar plots, were employed to illustrate and interpret the rich data. Our findings reveal important insights into the evolution of new cases over time, the distribution of vaccination rates across continents, the impact of demographic factors on COVID-19-related deaths, and the effectiveness of stringency measures implemented by authorities. Surprisingly, the analysis highlighted variations in mortality rates among different age groups, suggesting a potential disparity in pandemic response. The study underscores the importance of collective responsibility in combating viral outbreaks, and emphasizes the need for targeted interventions based on demographic considerations. Overall, this paper contributes to a better understanding of the COVID-19 pandemic through graphically informative data visualizations.

Index Terms—data analysis, data visualization, epidemic outbreak, COVID-19.

I. INTRODUCTION

The COVID-19 pandemic has undeniably marked one of the most transformative and challenging periods in recent history [1]–[3]. At the time of this study, COVID-19 is no longer a prevalent and widely discussed topic. Although it continues to exist, preventive measures have been implemented and both social media and the authorities have reduced their focus on the subject. The battle against this pandemic has been a long one, beginning with the first reported case in Wuhan, China, on December 1, 2019. Within a few months, the virus spread rapidly across the globe, causing a significant global impact. During this period, scientific advances in computational epidemiology have provided innovative tools to track and control epidemic outbreaks [4]–[7].

COVID-19 is caused by the SARS-CoV-2 virus, primarily transmitted through respiratory droplets from infected individuals. It can also spread by touching contaminated surfaces and then touching the face [8]. Symptoms vary widely, ranging from mild cold or flu-like symptoms to severe cases requiring hospitalization or leading to death. Some infected individuals may show no symptoms, complicating the determination of transmission rates. Initial estimates of transmission rates may have underestimated the actual value due to these factors [8].

To combat the rapid spread of COVID-19, authorities implemented various preventive measures. Lockdowns, social

distancing, closure of non-essential businesses, and mandatory use of face masks were among the main non-pharmaceutical interventions (NPIs) employed to reduce contact and limit virus transmission. Travel restrictions and quarantine measures were also implemented to prevent the introduction and spread of the virus. In healthcare settings, dedicated COVID-19 units were established to provide specialized care while minimizing the risk of transmission [9].

In addition to these preventive measures, the development of multiple vaccines has been a crucial milestone in the fight against COVID-19. Rigorously tested and proven effective, these vaccines have been approved for use worldwide. Vaccination campaigns prioritize high-risk groups with the goal of equitable distribution [10]. Vaccines play a critical role in controlling the spread of the virus and safeguarding public health.

As the world continues to navigate the challenges posed by COVID-19, collective efforts of preventive measures and vaccination hold the key to mitigating the impact of this global health crisis. By remaining vigilant, informed, and united, we can overcome challenges and strive towards a healthier and more resilient future.

The main focus of this study is the analysis of the COVID-19 pandemic, which has played a significant role in global health and society. It presents the evolution of the pandemic from its initial outbreak, highlighting the rapid spread of the virus and its global impact. In addition, we explore preventive measures, vaccination efforts, and the role of data visualization in understanding and combating the pandemic, underscoring the importance of collective efforts in mitigating its effects and ensuring a healthier future.

Data visualization is a key analysis technique employed in Big Data. Similar studies have focused on providing rich self-explanatory maps of the progression of COVID-19 over time, such as the work of Leung et al. [11]; the authors present choropleth and bubble maps of infectious hotspots across the globe. Authors Zhang et al. [12] describe the design of a pandemic dashboard with rich available visualizations. Authors Pang et al. [13] present alterations of COVID-19, in time, in the context of continents and countries. Authors Rupani et al. [14] provide visualizations for the environmental impacts of the COVID-19 pandemic in terms of air pollution, waste management and air quality improvements. Thus, the

interest in understanding the available Big Data produced—during and after this crisis—is of relevant scientific interest.

Our paper is structured in the following order: Section II presents the dataset used and the data processing. Section III describes the results in terms of different visualization techniques. Lastly, Section IV discusses the results and draws the main conclusions from the visual data analysis, while also drawing possible future research directions.

II. METHODS

In order to improve the understanding of the COVID-19 disease for the general public and researchers alike, we have undertaken the task of designing and developing a comprehensive big data visualization using publicly available data on the pandemic. This study focuses specifically on the epidemiological data for COVID-19. In this section, we will present our concept and approach to effectively illustrate the intricacies of COVID-19 epidemiological data.

A. Data overview

The dataset *COVID-19 by Our World in Data* [15], [16] provides comprehensive information related to the progression of the coronavirus pandemic for 242 countries, every day, throughout the pandemic period (1225 days). In this study, we used a snapshot of the available data between 03.Jan.2020–05.Mar.2023. The information is segmented into various categories, including the number of confirmed cases, deaths, tests performed, vaccination rates, and other relevant indicators. The dataset is regularly updated and sourced from reliable international organizations, national health agencies, and research institutions.

To obtain a comprehension of the dataset, it is imperative to explore its constituents in detail. The dataset consists of 67 columns that cover a total of 308,064 entries, respectively 292,183 after data cleaning. These columns cover the primary data on confirmed cases, deaths, hospitalizations, and tests. The total population covered is 8,045,246,545, with 6,921,601 deaths, 765,222,135 positive cases (9.51%), 670,145,839 persons aged over 65 (8.32%), 2,913,060,941 smokers (36.2%), and 5,634,590,706 total vaccinations.

Given the large amount of data available, we have opted to carefully retain a specific subset of columns from the dataset. These selected columns (see Table I) have been chosen on the basis of their potential to generate intriguing visualizations and insights.

B. Data processing

Before proceeding further, data cleaning is needed on the data set. Our approach involves replacing all instances of NaN (missing) values with 0. This decision is primarily driven by our initial data review, which indicates that the majority of NaN values are a result of missing information rather than erroneous data in the respective rows.

Additionally, it has been observed that there are inaccurate data entries in the "location" and "continent" columns, where certain values do not align with their respective categories

TABLE I
COVID-19 DATASET WITH COLUMN EXPLANATIONS.

Parameter	Description
iso_code	ISO 3166-1 alpha-3—three-letter country codes.
continent	Continent of the geographical location.
date	Date of observation.
location	Geographical location.
new_cases	New confirmed cases of COVID-19. Counts can include probable cases, where reported.
total_cases	Total confirmed cases of COVID-19. Counts can include probable cases, where reported.
new_deaths	New deaths attributed to COVID-19. Counts can include probable deaths, where reported.
total_deaths	Total deaths attributed to COVID-19. Counts can include probable deaths, where reported.
total_vaccinations	Total number of vaccination doses administered.
people_vaccinated	Total number of people who received at least one vaccine dose.
people_fully_vaccinated	Total number of people who received all doses prescribed by the initial vaccination protocol.
population	Population.
aged_65_older	Share of the population that is 65 years and older.
median_age	Median age of the population.
female_smokers	Share of women who smoke.
male_smokers	Share of men who smoke.
stringency_index	Government response stringency index: composite measure based on 9 response indicators, rescaled to a value from 0 to 100.

(e.g., "low income" listed as a location representing countries worldwide). To ensure accurate visualizations, it is recommended to remove these incorrect data entries.

Furthermore, the "date" column is currently formatted as a *float*. To facilitate visualizations based on time, it is necessary to convert the "date" column to the *datetime64* data type. Lastly, note that the data set does not contain duplicate values.

III. RESULTS

In this section, we explore a various range of visualizations, showcasing the various techniques used to represent the data effectively. These visualizations include line charts, bar graphs, scatter plots, pie charts, etc. Each visualization is carefully designed to highlight specific aspects of the dataset and provide meaningful insights into research objectives.

A. Cases, Deaths and Vaccination: A geographical perspective

Geographical plotting, a notable method in data analysis and visualization, enables the display of data on a global scale or within specific regions, such as states within a country. This technique uses vivid and colorful representations to improve understanding and interpretation of data patterns and trends. By mapping the data to geographical regions, researchers and analysts can uncover spatial relationships, identify regional variations, and gain valuable information on the distribution and impact of phenomena across different locations [17].

We use geographical plotting to visualize and explore the evolution of new cases (see Figure 2a), new deaths (see Figure 2b) and vaccination rate (see Figure 1) at a moment during the mid-pandemic, in October 2021. By mapping the distribution of new cases/new deaths/vaccination rate across

different regions and countries, we gain a comprehensive understanding of the global impact of the virus. This visualization technique allows us to observe spatial patterns and trends in the spread of the disease, highlighting areas of high transmission rates and identifying hotspots. Through the use of color-coded maps and interactive features, we can visually track the progression of new cases/new deaths/vaccination rate over time. This not only provides a visual representation of the geographical distribution of the virus but also enables us to analyze the temporal variations and identify periods of significant growth or decline.

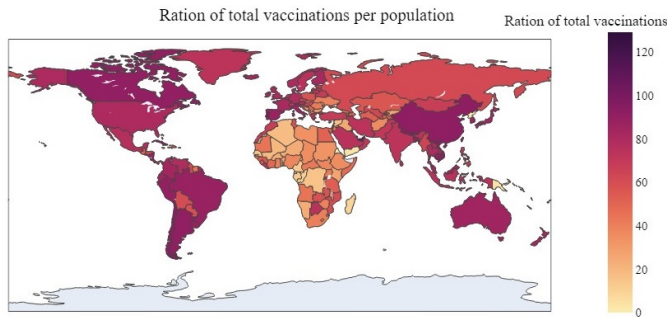


Fig. 1. Ratio of Total Vaccinations per Population: A visual representation showcasing the percentage of the population that has been vaccinated against COVID-19. This plot highlights the progress and coverage of vaccination efforts in different regions or countries, providing valuable insights into the global vaccination campaign.

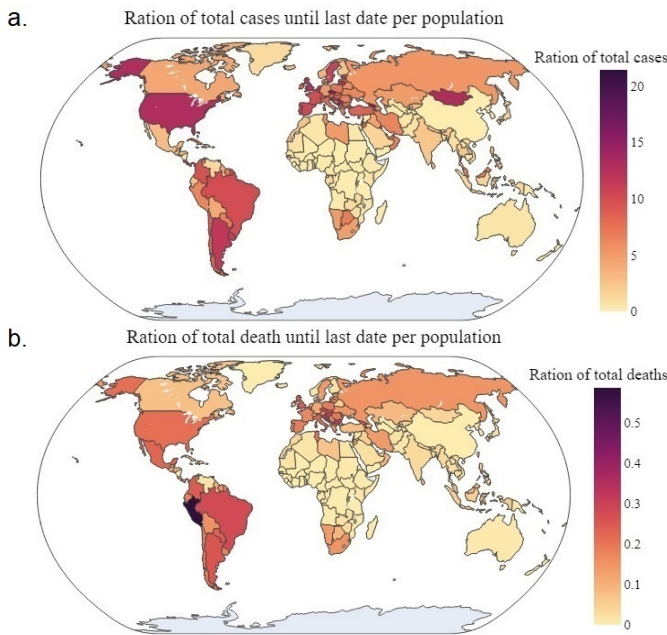


Fig. 2. (a) Ratio of total cases per population, and (b) ratio of total deaths per population, illustrating the proportion of confirmed COVID-19 cases in relation to the population size in October 2021.

B. Vaccination efforts across continents

Pie charts are commonly used to visually represent data by dividing them into slices of a circular pie. Each slice

represents a specific category or data point, with its size directly proportional to the quantity or percentage it represents. These charts are valuable for illustrating the composition or distribution of a whole dataset and highlighting the relative proportions of various categories or groups. Pie charts are appropriate for presenting categorical data with a limited number of distinct categories, offering a clear overview of the distribution and composition of the data [18].

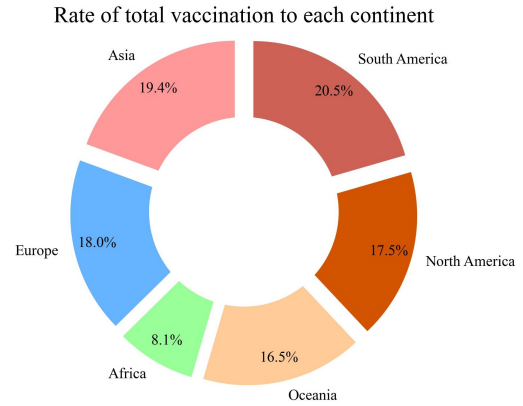


Fig. 3. Rate of vaccination by continent representing the varying rates of vaccination, allowing us to identify regions with higher or lower vaccination coverage. Understanding the vaccination rates by continent is crucial in assessing the global efforts to combat the pandemic and ensure widespread immunization.

This visualization is used to represent the COVID-19 vaccination rates (see Figure 3) across different continents. Each slice in the chart represents a specific continent, and its size is proportional to the percentage of the population that has been vaccinated in that category. These pie charts provide a valuable means of comparing and contrasting vaccination rates across continents.

C. Deaths–confirmed cases correlation

Scatter plots are a type of data visualization that displays the relationship between two numerical variables. They are constructed by plotting individual data points as dots on a graph with one variable represented on the x-axis and the other variable represented on the y-axis. This visualization is particularly useful for identifying patterns or relationships between variables, as they allow us to observe the distribution and concentration of data points. They can reveal trends, clusters, outliers, and the overall correlation between the two variables compared [19].

In Figure 4, the scatter plot serves as a valuable tool for understanding the impact of the pandemic by visualizing the correlation between the number of confirmed cases and the number of deaths in the countries with the highest death rates. It helps us identify any patterns or outliers, allowing for further analysis and comparison of the COVID-19 situation in different regions.

Alongside Figure 5, we provide valuable information on the relationship between the population aged 65 years and older and the number of deaths related to COVID. Presented as scatter plots, each data point is represented by a dot on

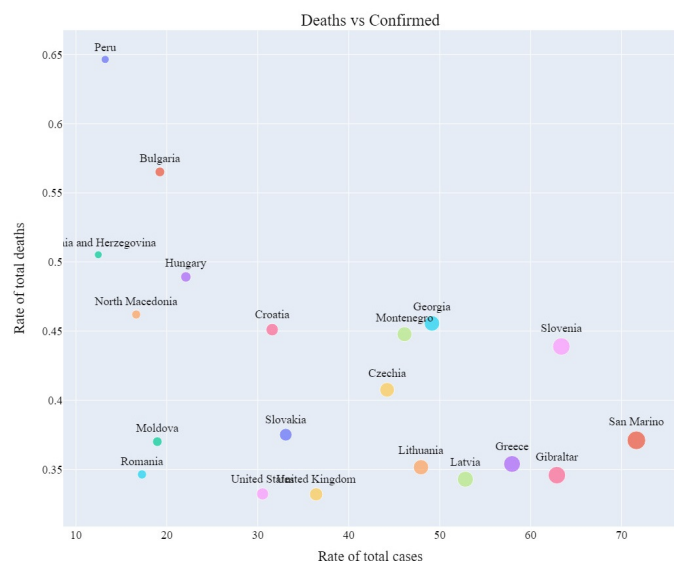


Fig. 4. Deaths vs. confirmed cases illustrating the relationship between the number of deaths and confirmed COVID-19 cases. The plot provides a visual representation of the correlation between the number of confirmed cases and deaths, allowing us to observe any trends or patterns.

the graph. The ox-axis denotes the population aged 65 and older, while the oy-axis represents the number of deaths. The visualization in panel Figure 5a offers a high-level view of the correlation between these two variables; it provides a general understanding of how the age group relates to the number of deaths. However, to gain a more comprehensive understanding, the second panel in Figure 5b offers a more in-depth analysis.

In Figure 6, we identify trends, patterns, or disparities between the two genders regarding the impact of smoking on COVID-19-related deaths. It serves as a useful tool for understanding the potential risks associated with smoking and its implications for individuals infected with the virus.

The scatter plot in Figure 7 depicts the impact of the stringency index (ox-axis) on the number of new cases of COVID-19 (oy-axis). This visualization allows us to examine the relationship between the level of stringency measures implemented by countries and the subsequent number of new cases reported.

D. Death rate distribution based on population age

Bar plots, also known as bar charts or bar graphs, are a common type of data visualization used to display categorical data. They consist of rectangular bars, where the length or height of each bar represents the value of a specific category or data point. One of the main advantages of bar plots is their simplicity and ease of interpretation. They allow for a clear visual comparison between different categories, making it easy to identify the largest or smallest values, as well as any significant differences or similarities [20].

This bar plot in Figure 8 provides a clear and concise representation of the relationship between median age and death rate. It allows for easy comparison between different age groups and facilitates the identification of any patterns or

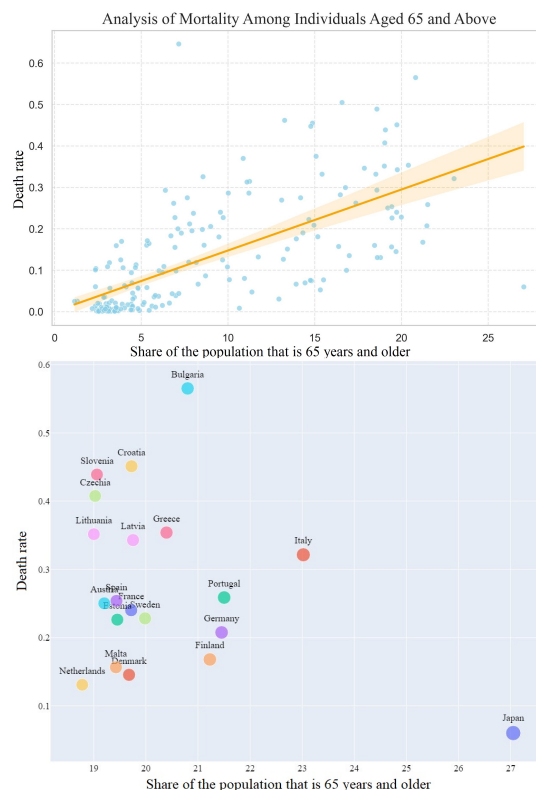


Fig. 5. Analysis of the mortality among individuals aged 65 and above representing the mortality rates among this specific demographic, providing valuable information on the severity of COVID-19 in relation to age.

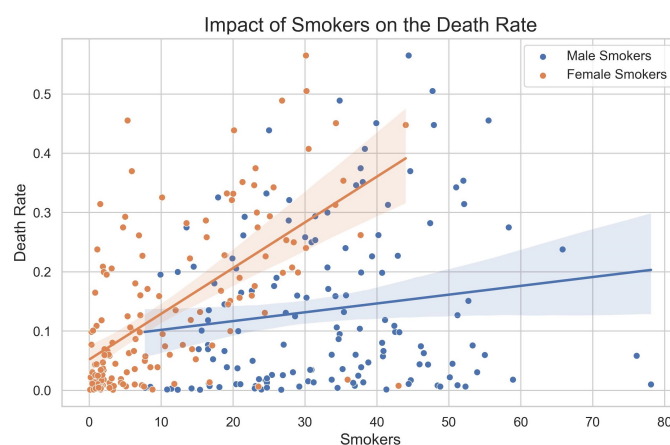


Fig. 6. The impact of smokers on the death rate providing a visual representation of the correlation between smoking rates and the corresponding death rates, shedding light on the association between these two factors.

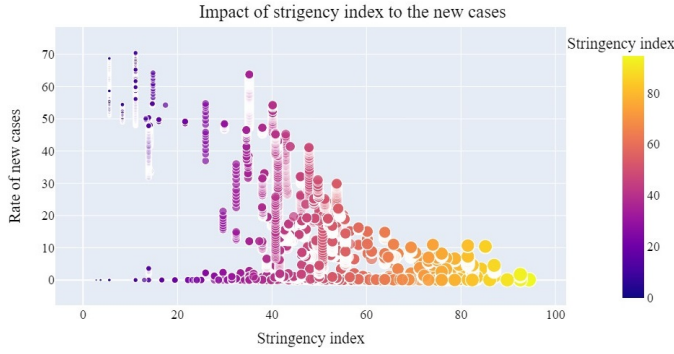


Fig. 7. The impact of the stringency index on new COVID-19 cases helps us understand the impact of various levels of stringency on the occurrence of COVID-19 infections and highlights the importance of implementing appropriate measures to contain the virus.

trends. By visually analyzing the bar plot, one can quickly assess the impact of age on the mortality rate and understand the vulnerability of different age groups to COVID-19 or other factors that contribute to the death rate.

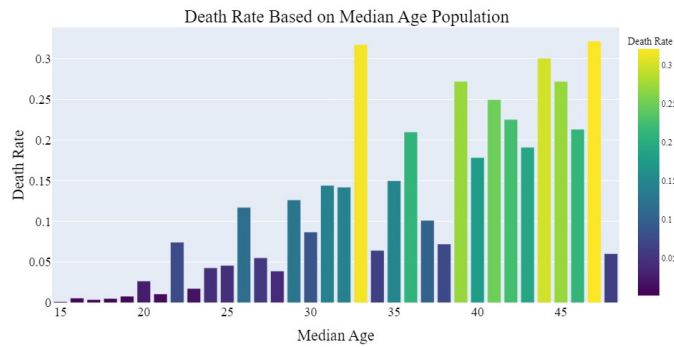


Fig. 8. Death rate based on the median population age providing valuable insights into the vulnerability of different age demographics to the virus and emphasizes the importance of considering age-related factors in pandemic response and healthcare strategies. Understanding the relationship between age and mortality can help inform targeted interventions and protective measures for at-risk populations.

IV. DISCUSSION AND CONCLUSIONS

Once the data have been pre-processed, we can embark on the exploration, analysis, and visualization process. Initially, our focus will be on understanding the global situation depicted by the dataset. Subsequently, we will progressively delve deeper into the analysis of the dataset, unraveling valuable insights at each stage.

A. Global situation assessment

To gain a comprehensive understanding of the global impact of the COVID-19 pandemic, it is crucial to monitor the daily progression of new cases, deaths, and vaccinated individuals relative to the total population in a specific region. This approach provides valuable information on the extent and implications of the pandemic on a global scale.

The evolution of new cases throughout the pandemic period: Despite the initial outbreak of COVID-19 occurring in China,

an analysis of the period from January 3, 2020, to April 24, 2023, reveals that China did not have the highest number of cases. Figure 2a illustrates that by October 8, 2021, both the United States and Europe had already exceeded a threshold of 10% new cases, whereas China had only 0.008% new cases. This prompts the important question of how China, along with several other countries, managed to maintain such a remarkably low infection rate.

The evolution of new deaths throughout the pandemic period: Similar observations can be made about the mortality rate (see Figure 2b). Although initial deaths were reported in China, the mortality rate remained relatively low throughout the entire duration of the pandemic. However, in countries such as Peru, Bulgaria, and others, the mortality rate exceeded 0.6%.

The proportion of vaccinated individuals in relation to the total population: Vaccination is a significant factor in controlling the spread of the virus. By examining the map, Figure 1, we can see that China and Argentina are among the countries with the highest vaccination rates, with approximately 91% of their population vaccinated. Following closely are Canada with a vaccination rate of 90% and Peru with 89%. In Europe, most countries have vaccination rates ranging between 70% and 80%.

The results of this study are expected to be of great interest to policy makers and public health officials, providing valuable information on the effective management of future outbreaks. Our findings can serve as a guide for decision makers in the development of policies and strategies to reduce the spread of epidemics. With the global threat of infectious diseases constantly evolving, it is increasingly important to understand the role of population density and mobility restrictions in controlling outbreaks.

B. Vaccination rate

To represent the vaccination rate in relation to the population of each continent, we opted to use a pie chart (see Figure 3). The chart reveals that the vaccination rate is comparatively consistent across continents, with the exception of Africa, where the rate is only 8.1%. The highest vaccination rate is observed in South America, reaching 20.5%.

C. Comparison between the number of confirmed cases and deaths

To track the mortality rate among confirmed cases of COVID-19, we analyzed the 20 countries with the highest death rates in relation to their population. Based on the analyzed data, shown in Figure 4, Peru had the highest rate, with 13.21% of infected individuals resulting in a mortality rate of 0.64%. This corresponds to approximately 4.8% of the diagnosed population.

D. The impact of age on COVID-related deaths

Based on Figures 5a–b, it is evident that there is a clear pattern, indicating that a significant portion of the variation in death rates can be attributed to the percentage of the population

aged 65 years or older. However, there is an outlier in the lower right corner, representing a country with a population of 27% in the age range of 65+. Interestingly, this outlier is Japan, known for its technological advancements and reputation for strong immunity and overall good health. It raises the question of whether Japan's unique factors contribute to its lower death rates among the elderly population.

E. The impact of smokers to the death rate

The impact of smoking on the death rate can be observed in Figure 6. This plot suggests that there is a notable correlation between smoking rates and death rates due to COVID-19. Countries with higher smoking rates tend to exhibit higher death rates. This trend indicates that smoking could be a contributing factor to the severity of the COVID-19 outcomes. Furthermore, it can be deduced that female smokers have a significantly higher susceptibility to mortality compared to their male counterparts.

F. Effectiveness of measures adopted during the pandemic

To evaluate the effectiveness of the measures implemented by the authorities, we analyzed the top 10 countries with the highest number of reported cases in relation to their population. The measures were quantified on a scale of 10 to 100, where a value of 0 represents no measures taken and 100 indicates drastic measures. Figure 7 clearly shows that when the stringency index is close to zero, the number of new cases experiences a significant increase. By contrast, as the stringency index increases, the number of cases decreases. This shows an inverse relationship between the stringency index and the number of cases, indicating that stricter measures are associated with lower case numbers.

In conclusion, the measures implemented by the authorities have yielded positive results by significantly reducing the number of new cases.

G. Death rate based on median age population

The visualization in Figure 8 reveals an intriguing finding, namely that the highest mortality rate is unexpectedly associated with individuals at the median age of 47 years, with the following highest mortality rate observed among those at the median age of 33 years. This unexpected pattern suggests that age alone may not be the sole determining factor for COVID-19-related deaths. However, despite the earlier analysis indicating that 65- and over-aged individuals are more vulnerable to COVID-19-related deaths, this visualization provides an alternative perspective on the pandemic response. It suggests that those at higher risk have taken significant precautions to protect themselves, while younger individuals, who may perceive themselves as less susceptible, may have overlooked the importance of protective measures. This observation prompts reflections on the varying levels of awareness and adherence to precautionary measures among different age groups. It underscores the importance of fostering a collective sense of responsibility to prioritize the safety and well-being of all individuals, regardless of their age.

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